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ScienceDirect
Chemical Geology 247 (2008) 401 – 418
CHEMICAL GEOLOGY
INCLUDING
ISOTOPE GEOSCIENCE
www.elsevier.com/locate/chemgeo

Origin of the inconsistent apparent Re–Os ages of the Jinchuan Ni–Cu sulfide ore deposit, China: Post-segregation diffusion of Os

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Received 8 April 2007; received in revised form 9 November 2007; accepted 13 November 2007

Editor: R.L. Rudnick

Abstract

Apparent Re–Os ages of some magmatic sulfide ore deposits are older than the zircon and baddeleyite U–Pb ages which are interpreted as the formation age of the host intrusions. The Jinchuan Ni–Cu–PGE deposit of China, the world's third largest, is such a case. We report apparent Re–Os isochron ages of 1117±67 Ma, 1074±120 Ma and 867±75 Ma with initial $^{187}Os/^{188}Os$ ratios of 0.120±0.012, 0.162±0.017 and 0.235±0.027 for disseminated ores, sulfides from the disseminated ores and massive ores from Jinchuan, respectively. Using these data and Re–Os ages from the literature, we find that the oldest apparent Re–Os age and lowest initial Os isotope ratio are from disseminated ores which contain small amounts of sulfide minerals, the highest initial Os isotope ratios and youngest apparent Re–Os ages, consistent with the zircon and baddeleyite U–Pb ages, are from massive ores containing 90–100 modal% sulfide, and net-textured ores with about 25 modal% sulfides yield apparent Re–Os ages and initial Os ratios intermediate between those of the disseminated and massive ores.

Because Os diffusion between sulfides is inhibited by the intervening silicates even at high temperatures, re-equilibration did not occur in the disseminated ore and the samples retained the Os ratios of the contaminated magma, leading to geologically meaningless ages that are older than the formation age of the rocks. While Os-bearing sulfide minerals and magnetite show low closure temperatures of Os diffusion and the sulfide minerals in the massive ore are closely connected with each other, facilitating fast diffusion of Os, re-equilibration of Os was achieved during cooling of the ore from about 850 °C after the segregation to about 400 °C. Thus, an age corresponding to the formation time and an elevated initial Os ratio were yielded by the massive ore. Os isotopes in the net-textured ore behave in the way intermediate between the disseminated and massive ores. Pb isotope data support the Os results. Disseminated ores have heterogeneous Pb isotope ratios whereas Pb in the massive ores is more uniform, consistent with Pb isotopic equilibration in the massive ores, but not in the disseminated ores.

Keywords: Apparent Re–Os isochron age; Apparent initial Os isotopic ratio; Diffusion of Os; Jinchuan Ni–Cu–PGE deposit; China

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0009-2541/$ - see front matter
doi:10.1016/j.chemgeo.2007.11.002

1. Introduction

Magmatic sulfide Ni–Cu ore deposits are important sources of Ni and, to a less extent, Cu and platinum group elements (PGE). Precise determination of the formation age of these deposits is crucial to understanding the processes of mineralization. Re–Os isotopic geochronology has been proven to be a powerful tool for dating various types of mineralization, including Ni–Cu sulfide deposits (e.g. Shirey and Walker, 1998). For instance, the Noril'sk I and Talnakh intrusions yield a Re–Os age of 241.9 ± 0.6 Ma and the Kharaelakh intrusion an age of 243.1 ± 3.8 Ma (Walker et al., 1994) (the ^{187}Re decay constant of 1.666×10^{-11} y^{-1} is used throughout this paper, and all old data have been re-calculated using this value; Smoliar et al., 1996), which agree very well with the SHRIMP zircon age of 248 ± 4 Ma for the Noril'sk I intrusion (Campbell et al., 1992) if the small uncertainties in the decay constants are taken into account (Begemann et al., 2001). Undisturbed sulfide ore samples of the Kambalda Ni–Cu–PGE deposit hosted by komatiite in Western Australia yield a Re–Os age of 2664 ± 36 Ma. This age is close to the U–Pb zircon ages of 2703 ± 4 Ma and 2709 ± 6 Ma for sediments within the komatiite pile (Forster et al., 1996).

However, Re–Os ages of the magmatic sulfide ore deposits do not always agree with the zircon and baddeleyite U–Pb ages of the host mafic intrusions and, in some cases, are younger than the host. Precise U–Pb dating of coexisting baddeleyite and zircon from the Voisey's Bay Ni–Cu–Co deposit of Labrador, Canada indicates that all the mafic rocks of the intrusion were emplaced at 1332.7 ± 1.0 Ma (Amelin et al., 1999). Large whole-rock sulfide samples yield an imprecise Re–Os isochron age of about 1302 Ma that is consistent with the baddeleyite U–Pb ages, but, mineral separates give an isochron age of about 988 Ma. These dates were interpreted to be the result of resetting of the Re–Os isotopic system at the mineral scale during a later thermo-hydrothermal event that coincided with the Grenville orogeny (Lambert et al., 2000).

On the other hand, some magmatic ore deposits show Re–Os dates older than the zircon and baddeleyite U–Pb and/or Sm–Nd ages for intrusions. The Stillwater Complex yields a U–Pb zircon–baddeleyite age of 2705 ± 4 Ma (Premo et al., 1990), which agrees well with the Sm–Nd mineral isochron age of 2701 ± 8 Ma (DePaolo and Wasserburg, 1979). However, chromite separates from unit H of the Complex yield a Re–Os isochron age of 2904 ± 43 Ma, significantly older than the zircon–baddeleyite U–Pb age (Horan et al., 2001). Re–Os data do not form reasonable Re–Os isochrons for some other deposits, such as in Sally Malay intrusion, East Kimberley, Western Australia (Sproule et al., 1999) and the Duluth Complex, Minnesota, USA (Ripley et al., 1998). Assimilation of the crust and variations in the *R* factor were invoked to explain those imprecise isochrons or errorchrons.

The Jinchuan sulfide Ni–Cu–PGE ore deposit located in northwestern China is another example in which inconsistent apparent Re–Os ages have been obtained for different types of rocks and ores. Reported Re–Os ages for this deposit range from 1408 ± 34 Ma for net-textured ore to 853 ± 25 Ma for massive ore (Keays et al., 2004; Zhang et al., 2004; Yan et al., 2005; Yang, 2005; Yang et al., 2005; Yang, 2006). If ages yielded by other dating methods are included, the apparent age span becomes even larger. For example, Sm–Nd data yield an isochron age of 1508 ± 31 Ma for the mafic rocks (Tang et al., 1992). Recently reported zircon and baddeleyite U–Pb ages of about 830 Ma (Li et al., 2004, 2005) are taken as the formation age of the mafic intrusion and associated sulfide ore deposit. Therefore, the deposit offers an excellent opportunity to investigate the causes of the inconsistent Re–Os apparent dates.

In this paper, we report Re–Os ages for disseminated ore and sulfide separate from disseminated ore in the Jinchuan deposit for the first time. We also present one more Re–Os age for massive ore in Jinchuan to check if massive ores collected from different parts of the deposit give the same age. Based on our new dates, together with previous dating results, we develop a model that explains the controversial ages for this deposit. We then test the applicability of this model to other magmatic sulfide deposits in the world. Pb isotope ratios for disseminated and massive ores from the Jinchuan deposit are reported to support our model.

2. Geological setting and samples

The Jinchuan ultramafic intrusion in northwest China has reserves of nearly 500 million tones of Ni–Cu sulfide ore (grading 1.2% Ni and 0.7% Cu) and is ranked the world's third largest magmatic Ni–Cu sulfide deposit following Sudbury and Noril'sk (Naldrett, 1999). The deposit is also one of the largest PGE deposits of China (Yang et al., 1993). It is characteristic that the huge Jinchuan Ni–Cu sulfide deposit is formed in a small intrusion.

The Jinchuan intrusion is located in the Longshoushan Uplift, a fault-bounded, northwest-striking terrane between the Alax massif (the westernmost part of the North China craton) to the north and the northern Qilian–

Qinling Orogenic Belt to the south (Fig. 1a). It intruded Paleoproterozoic migmatite, gneiss, and serpentine marble of the Longshoushan Group and these rocks have been metamorphosed to the high amphibolite facies. The intrusion itself is cut by dolerite dykes. The intrusion, as a dyke-like body, is about 6500 m long and 20 to 527 m wide, and has a total exposure area of about 1.34 km^2. The intrusion and its sulfide ores are divided into four mining areas (from west to east, III, I, II and IV) by NE-trending faults F8, F16-1 and F23 (Fig. 1b, c; S.G.U., 1984; Chai and Naldrett, 1992; Tang and Li, 1995).

The main rock types include dunite, lherzolite, plagioclase lherzolite, troctolite and olivine pyroxenite (S.G. U., 1984; Chai and Naldrett, 1992; Tang and Li, 1995). The contacts between different rock types are mostly gradational (Chai and Naldrett, 1992), but clear divisions can also be observed in the field (Tang and Li, 1995). Generally, the intrusion features a dunite core, lherzolite and plagioclase lherzolite wings, and olivine pyroxenite margins. The dunite core is commonly closer to the footwall side of the intrusion. However, in the eastern part of the intrusion there is an upward succession from dunite at the base to lherzolite and plagioclase lherzolite at the shallower levels. The net-textured sulfide ores containing 8–30% Ni–Cu sulfide are commonly hosted by dunite, and to lesser degree, by lherzolite. The net-textured ores are commonly surrounded by disseminated ores, which are hosted in lherzolite and plagioclase lherzolite. The massive ores with sparse silicate minerals are veins that cut the net-textured ores near the footwall side of the intrusion (Fig. 1; S.G.U., 1984; Chai and Naldrett, 1992; Tang and Li, 1995).

All samples were collected from No. 1 ore body at the mining area II (Fig. 1b,c).

Disseminated ore samples (04JCII-035 to 043, short as JCII-035 to 043) were collected from the northwest part of the ore body 1 in a shaft along the drill grid 12 at the 1118 m level near the footwall direction of the intrusion

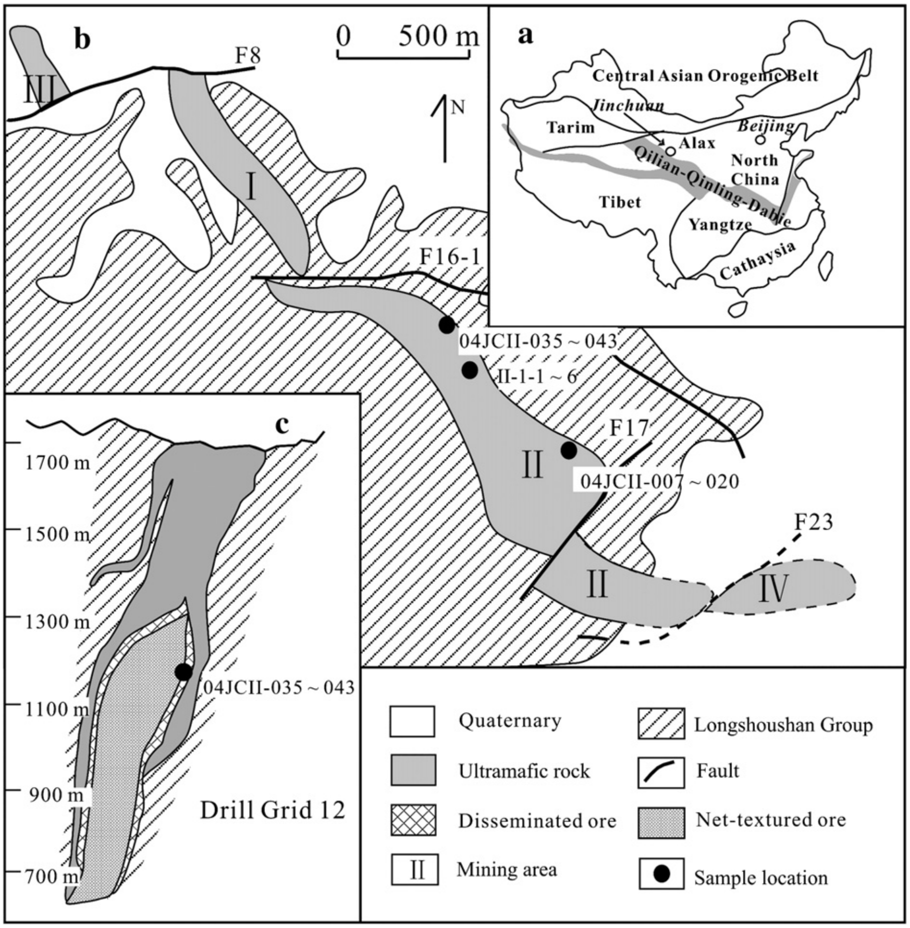

Fig. 1. a) The location of Jinchuan (modified after Li et al., 2005); b) the geologic sketch map of the Jinchuan Ni–Cu sulfide ore deposit (modified after Tang and Barnes, 1998); c) the cross section of the drill grid 12 (modified after Yang et al., 2006). Sample locations are shown in parts of b and c.

(Fig. 1). The samples were collected from a limited area in order to minimize possible spatial heterogeneity in initial Os isotopic ratio. The host rock is plagioclase lherzolite and consists of olivine (60–70 modal%), pyroxene (20–30 modal%, mostly clinopyroxene), plagioclase (5–10 modal%), and a little phlogopite in places. Olivine mostly occurs as subhedral crystal (1–3 mm) and sometimes as inclusion in the pyroxene. Small orthopyroxene and chrome spinel crystals occur as inclusion in olivine. Pyroxene (1–5 mm) and plagioclase (0.5–2 mm) are anhedral and occur interstitially. The rock was altered to different degrees. The olivine is replaced by serpentine and very fine-grained magnetite along fractures, the pyroxene is rimmed by serpentine, chlorite and uralite and the plagioclase is strongly sericitized (Fig. 2b,c). The rock contains less than 2 modal% opaque minerals (pentlandite, pyrrhotite, chalcopyrite and magnetite) which occur interstitially.

Disseminated ore samples (II-1-1, 2, 5 and 6) were collected from shafts along the drill grids 20 to 22 at the 1198 m level (Fig. 1b). These samples are similar to samples JCII-035 to 043 in the lithology, but contain more sulfide minerals (6–8 modal%) and show disseminated and semi-net-textured structures. Sulfides were separated from the ore by floatation and magnetic separation. Sulfide separates contain more than 96% sulfides, including pentlandite, pyrrhotite, chalcopyrite and magnetite. Subscripts "s" and "m" represent slightly different magnetic field applied and the "s" series contains about 5% more chalcopyrite than the "m" series.

The massive ore samples (04JCII-007 to 020, short as JCII-007 to 020) were collected at 1118 m level, but, near the east part of the body and the footwall side of the intrusion (Fig. 1b). They consist of pentlandite (10–20 modal%, 0.1–0.5 mm), pyrrhotite (65–80 modal%,

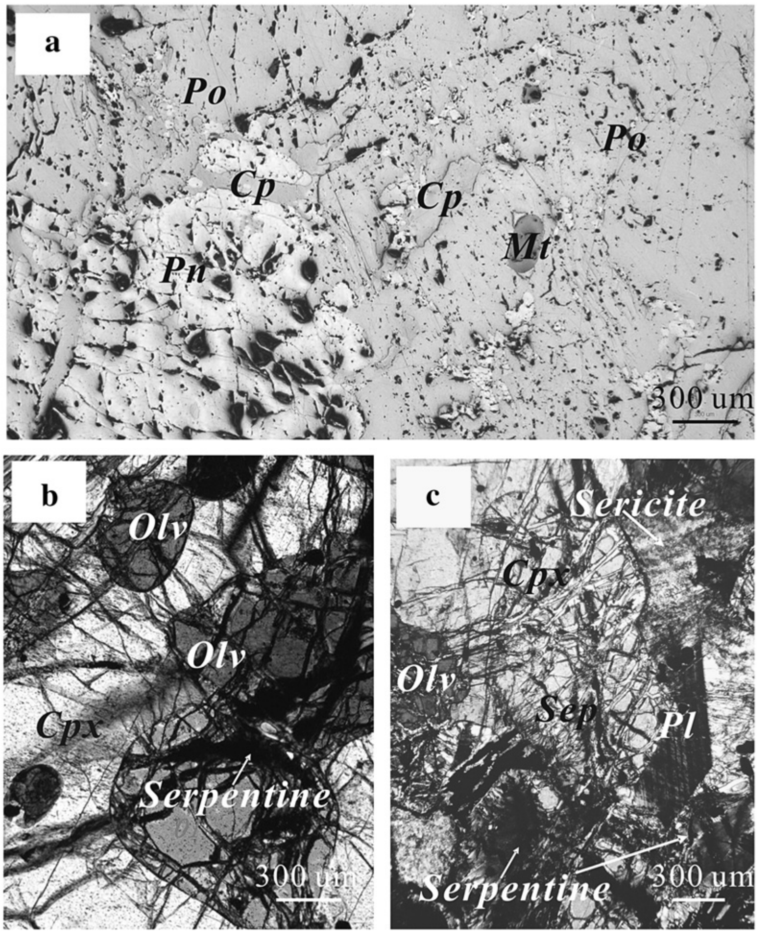

Fig. 2. Photomicrographs of the rock and ore in Jinchuan. (a) massive ore (reflected, plane-polarized light, sample JCII-008); b) less altered plagioclase lherzolite (cross-polarized light, sample JCII-43); c) strongly altered plagioclase lherzolite (cross-polarized light, sample JCII-40).

0.1–0.5 mm, sometimes up to 2 mm), chalcopyrite (3–10 modal%, 0.01–0.5 mm), pyrite (1–5 modal%, 0.01–0.05 mm), minor magnetite (0.1–0.6 mm) and limonite as well as trace silicate minerals. These minerals all occur as anhedral crystals (Fig. 2a). Sometimes, pentlandite occurs as exsolved lamellae in pyrrhotite. Veinlet of pyrite often cuts through pentlandite and pyrrhotite. Chalcopyrite replaces pentlandite and pyrrhotite. Magnetite is replaced by sulfide minerals.

3. Analytical methods

3.1. Re–Os isotopes

All Re–Os chemistry and Inductively Coupled Plasma Mass-spectrometric measurements (ICP-MS) were performed in the National Research Center of Geoanalysis (NRCG), Chinese Academy of Geological Sciences. Re and Os concentrations were determined by isotope dilution. The ^{185}Re and ^{190}Os spikes were purchased from Oak Ridge National Laboratory, USA, and calibrated at the NRCG. The samples were crushed and grounded to less than 200 mesh using contamination-free equipment. About 2 g of disseminated ore and 0.2 g of sulfide and massive ore were weighed precisely and loaded into Carius tubes, together with ^{185}Re and ^{190}Os mixed spikes and digested by reverse aqua regia by presence of small amount of H_2O_2 (Shirey and Walker, 1995). The tube was heated in an oven at 230 °C for about 24 h. Os was separated as OsO_4 by distillation at 105–110 °C and was trapped by Milli-Q water. During distillation, the reverse aqua regia was acted as the oxidizer. Re was extracted from the residue by acetone in 5 M NaOH solution (Du et al., 2001, 2004).

Re concentrations of all samples and Os concentrations and isotopic compositions of sulfides from disseminated ores and massive ore samples were measured by ICP-MS (PQ Excell). The intensity of the ^{190}Os signal was monitored to correct for traces of Os in Re and ^{185}Re intensity was used to monitor traces of Re in Os. H_2O_2 and 5% ammonia were repeatedly used to wash the Teflon injection tube in order to avoid cross-contamination between different samples (Du et al., 2001, 2004).

Os concentrations and isotopic compositions of the disseminated ore samples were measured by a negative thermal ionization mass-spectrometer (N-TIMS, MAT-262) at the CAS Key Laboratory of Crust-Mantle Materials and Environments, University of Science and Technology of China (LCMME), because of the low Os concentrations in these samples. Before each N-TIMS measurement, the Os solutions were further purified by micro-distillation (Birck et al., 1997). The intensity of the mass 233 peak was monitored to correct for traces of Re in Os. M233/M236 ratios (approximately equivalent to ^{185}Re/^{188}Os ratios) for all samples were less than 0.00002. All measured ratios were corrected for oxygen isotopic compositions (Sun et al., 1997) and iteration was used to correct mass fractionation and to calculate Os concentrations and isotopic compositions (Yang, 2005; Yang, 2006). The total procedural blanks were about 5 pg for Re and 1 to 0.2 pg for Os based on blank runs analyzed together with the samples. The ^{187}Os/^{188}Os of the blank is 0.25 based on routine measurements of blanks. The Certified Material GBW04436 (JDC, a molybdenite from the Jinduicheng

Table 1
Re and Os isotopic data for disseminated and massive copper-nickel sulfide ores from Jinchuan

	Re (ppb)	Os (ppb)	^{187}Re/^{188}Os	^{187}Os/^{188}Os
Disseminated ore (102°09′ 59″, 38°28′47″ to 102°09′58″, 38°28′46″)				
04JCII-035	5.084	1.891	12.86	0.3650
04JCII-035	5.084	1.881	12.93	0.3657
04JCII-035	5.047	1.892	12.76	0.3650
04JCII-036	0.663	0.313	10.19	0.3067
04JCII-037	0.866	0.332	12.50	0.3542
04JCII-038	0.698	0.466	7.160	0.2586
04JCII-039	0.938	0.374	12.00	0.3421
04JCII-040	0.926	0.432	10.32	0.3101
04JCII-041	0.798	0.334	11.41	0.3363
04JCII-042	0.732	0.338	10.37	0.3093
04JCII-043	0.656	0.331	9.480	0.2947
Sulfides of the disseminated ore (102°10′09″, 38°28′36″ to 102°10′11″, 38°28′34″)				
II-1-1sa	137.8	78.70	8.417	0.3133
II-1-1sb	133.0	79.06	8.088	0.3103
II-1-2s	195.5	121.3	7.751	0.2940
II-1-5s	121.4	48.93	11.93	0.3737
II-1-6s	222.7	210.0	5.101	0.2579
II-1-1ma	177.2	105.4	8.039	0.3129
II-1-1mb	179.6	105.7	8.131	0.3106
II-1-2m	213.9	137.0	7.467	0.2885
II-1-5m	147.2	65.16	10.80	0.3632
II-1-6m	255.2	232.2	5.256	0.2571
Massive ore (102°10′28″, 38°28′17″)				
04JCII-007	504.7	129.7	18.72	0.5027
04JCII-008	387.8	75.4	24.72	0.5889
04JCII-009	472.9	115.5	19.68	0.5230
04JCII-010	505.1	128.2	18.94	0.5088
04JCII-011	374.2	78.9	22.71	0.5641
04JCII-012	529.3	130.0	19.58	0.5203
04JCII-013	431.5	89.1	23.29	0.5709
04JCII-014	435.3	101.4	20.52	0.5332
04JCII-015	356.2	67.45	25.25	0.6046
04JCII-016	427.8	101.8	20.10	0.5407
04JCII-020	362.4	71.67	24.19	0.5837

Errors of 5% for ^{187}Re/^{188}Os and of 1% for ^{187}Os/^{188}Os of samples determined by N-TIMS and of 2% for ^{187}Os/^{188}Os of samples determined by ICP-MS are assigned.

deposit, China) was used to monitor the accuracy of the measurements. The Re and Os contents and Re–Os ages of JDC molybdenite determined during the course of this work are 17,336±41 ppb, 25.25±0.07 ppb and 138.9±0.5 Ma (n=17), respectively, which coincide within uncertainties with the reference value (Re= 17,390±320 ppb, Os=25.46±0.60 ppb and t=139.6 ±3.8 Ma; Du et al., 2004).

Replicate analyses (3 of JCII-035, 2 of II-1-1s and 2 of II-1-1 m) were performed. All results are in agreement or plot on the same regression line (Table 1; Fig. 3), suggesting the reliability of our analyses.

3.2. Pb isotopes

Pb isotopic analyses were performed at the LCMME. Ore samples were first dissolved in 6 M HCl+concentrated HNO_3 for 24 h to obtain the sulfide phases. The residue was washed by Milli Q water and then dissolved by HF+HNO_3 for another 24 h to obtain the silicate phases. Both sulfide and silicate aliquots were converted to 1.5 M HCl+0.65 M HBr solutions. Pb was separated by chromatography on Bio-Rad AG-1×8 anion exchange resin and eluted by 1.0 M HNO_3. Pb isotopic ratios were determined on a Finnigan MAT-262 multi-collector thermal ionization mass spectrometer in static mode. Measured isotopic ratios were corrected for a mass fractionation of 0.10% per atomic mass unit determined by replicate measurements of reference sample SRM 981. Replicate analyses of RSM 981 (recommended values are: $^{206}Pb/^{204}Pb$=16.937, $^{207}Pb/^{204}Pb$=15.491 and $^{208}Pb/^{204}Pb$=36.721) gave $^{206}Pb/^{204}Pb$=16.903±0.011, $^{207}Pb/^{204}Pb$=15.446±0.013, and $^{208}Pb/^{204}Pb$=36.566± 0.041 (N=31, 2σ).

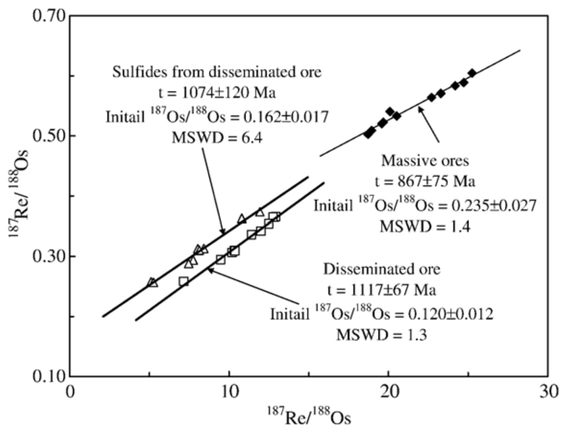

Fig. 3. Re–Os isochrons for disseminated ore, sulfides from disseminated ore and massive ore from Jinchuan.

4. Analytical results

4.1. Re–Os isotopes

Re and Os concentrations and $^{187}Re/^{188}Os$, $^{187}Os/^{188}Os$ ratios from Jinchuan, corrected for total blanks, for disseminated and massive ores as well as sulfides from disseminated ores, are listed in Table 1. Errors of 5% for $^{187}Re/^{188}Os$ and of 1% for $^{187}Os/^{188}Os$ for samples determined by N-TIMS and of 2% for $^{187}Os/^{188}Os$ for samples determined by ICP-MS are assigned. The assigned 5% error in $^{187}Re/^{188}Os$ includes uncertainties in sample and spike weighing, uncertainties in spike calibration, difference of our standard solutions and the one used for determining decay constant, uncertainties in mass-spectrometry (both in run precision and that in fractionation correction by normalization). The assignment of the 1% and 2% errors in $^{187}Os/^{188}Os$ ratio is based on replicate analyses of the laboratory reference samples and includes uncertainties in mass-spectrometry, both in run precision and that in fractionation correction by normalization. The Re–Os isochrons calculated by ISOPLOT (Ludwig, 2004) are shown in Fig. 3.

The disseminated ore samples have Re and Os concentrations ranging from 0.656 to 5.084 ppb and 0.313 to 1.892 ppb, respectively, and yield an apparent Re–Os isochron age of 1117±67 Ma with an apparent initial $^{187}Os/^{188}Os$ ratio of 0.120±0.012 (n=11, MSWD=1.3; Fig. 3). Large uncertainties in the isochron age and initial isotopic ratio are probably caused by small variation in $^{187}Re/^{188}Os$ ratios (7.16–12.93). Small variations in the $^{187}Re/^{188}Os$ and $^{187}Os/^{188}Os$ are possibly resulted from that samples were collected within a very small area. The age determined here differs from previously reported Re–Os ages of the net-textured and massive ores (Keays et al., 2004; Zhang et al., 2004; Yang, 2005; Yan et al., 2005; Yang et al., 2005), as well as the zircon and baddeleyite U–Pb ages (Li et al., 2004, 2005; Yan et al., 2005).

Although the Jinchuan intrusion is characterized by a complex history of hydrothermal alteration and our disseminated ore samples are a mixture of silicates (including rock-forming and altered minerals), sulfides and oxide, whether these minerals possess different initial isotopic compositions is not known. In addition, petrologic studies did not tell whether the formation and alteration ages are distinguishable by the present (Re–Os) geochronometer. However, Tang and Barnes (1998) suggested that volatile content of the magma increased with decreasing temperature and continued crystallization and that it is those volatiles that resulted in autometamorphism, forming serpentine, amphibole and chlorite after olivine and pyroxene. This infers that the main stage of the alteration was closely after the crystallization of rock-forming and sulfide minerals.

In this case, if rock (or ore) system and the minerals included have a homogeneous initial isotopic ratio and the rock remained in a closed system after the alteration, the Re–Os isochron constructed by the unaltered and altered samples would both show good linearity and yield the same age and initial isotopic ratio. Suppose that the initial state of the isotopic ratios of the rock body is a mixing line with positive or negative slopes, even if the alteration took place shortly after the formation of the rock and the rock remained in a closed system after alteration, the initial state of the $^{187}Re/^{188}Os$–$^{187}Os/^{188}Os$ relation would very likely not keep lineally correlated because of the shift of the $^{187}Re/^{188}Os$ ratios, thus, “the Re–Os isochron” constructed by altered samples would show poor linearity and yield an apparent age and initial isotopic ratio which are deviate from that given by the isochron formed by the unaltered minerals (Fig. 4).

Our data of the disseminated ore show good linearity that suggests homogeneous or well lineally correlated initial Os isotopic ratios and infer that no significant shift in $^{187}Re/^{188}Os$ ratios occurred.

The sulfide samples from disseminated ore show Re and Os concentrations ranging from 121.4 to 255.2 ppb and 48.93 to 232.2 ppb, respectively, and yield an model 3 isochron age of 1074±120 Ma with an apparent initial $^{187}Os/^{188}Os$ ratio of 0.162±0.017 (n=10, MSWD=6.4; Fig. 3). This apparent Re–Os isochron age of 1074±120 Ma agrees within analytical uncertainties with that of the whole rocks and the initial $^{187}Os/^{188}Os$ ratio of 0.162±0.017 is slightly higher than that given by the whole rock.

Poor linearity, Model 3 age and large uncertainties in the isochron age and initial isotopic ratio are caused not

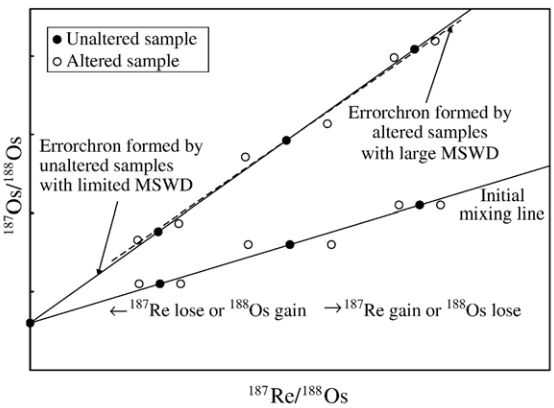

Fig. 4. A $^{187}Re/^{188}Os$ vs. $^{187}Os/^{188}Os$ plot, showing the effect of the shift in $^{187}Re/^{188}Os$ caused by alteration on the Re–Os system with an initial state of being a mixing line.

only by small variation in $^{187}Re/^{188}Os$ ratios (5.101–11.93) but also by heterogeneous or not well correlated initial Os isotopic ratios due to large sampling area and spatial heterogeneity in initial Os ratios. The high initial Os ratio is resulted from higher sulfide content (6–8 modal %) in these samples (see Sections 5.2, 5.4).

A rough comparison of Re and Os concentrations in the disseminated ores and sulfides suggests that Os is dominant by sulfides in the disseminated ore and that the isochron of the whole rock of the disseminated ore samples is in fact controlled by the Os in sulfides, but not an artifact of mixing altered and primary silicates and sulfides.

The massive ore samples have Re and Os concentrations ranging from 356 to 529 ppb and 67 to 130 ppb, respectively, and yield an apparent Re–Os isochron age of 867±75 Ma with an apparent initial $^{187}Os/^{188}Os$ ratio of 0.235±0.027 (n=11, MSWD=1.4; Fig. 3). Large uncertainties in the isochron age and initial ratio are possibly caused by small variation in $^{187}Re/^{188}Os$ ratios (18.72–25.25) due to small sampling area. The age agrees within uncertainties with previously reported Re–Os ages of the massive ores (Yang, 2005; Yang et al., 2005), and the zircon and baddeleyite U–Pb ages (Li et al., 2004, 2005; Yan et al., 2005).

4.2. Pb isotopes

The disseminated ores have heterogeneous Pb isotopic ratios with $^{206}Pb/^{204}Pb$=16.826 to 17.314, $^{207}Pb/^{204}Pb$=15.367 to 15.465 and $^{208}Pb/^{204}Pb$=37.085 to 38.342 (Table 2). On the $^{206}Pb/^{204}Pb$ vs. $^{207}Pb/^{204}Pb$ diagram, these data plot close to the mantle evolution curve (Fig. 5a; Zartman and Doe, 1981) and form a linear array. On the $^{206}Pb/^{204}Pb$ vs. $^{208}Pb/^{204}Pb$ diagram, the data are spread between the lower crust and the other three (the mantle, orogenic and upper crust) evolution curves (Fig. 5b; Zartman and Doe, 1981) with roughly one end towards the lower crust. The massive ores have relatively homogeneous Pb isotopic ratios with $^{206}Pb/^{204}Pb$=16.742 to 16.945, $^{207}Pb/^{204}Pb$=15.357 to 15.418 and $^{208}Pb/^{204}Pb$=37.083 to 37.283 (Table 2). On the Pb-isotope evolution diagrams they plot close to the mantle curve (Fig. 5a), between the lower crust and the other three evolution curves (Fig. 5b) and coincide with the low end of the disseminated ore trends (Fig. 5).

5. Discussion

5.1. The formation age of the Jinchuan Ni–Cu deposit

Li et al. (2005) reported a SHRIMP U–Pb age of 812±26 Ma for baddeleyite from the plagioclase lherzolite, and

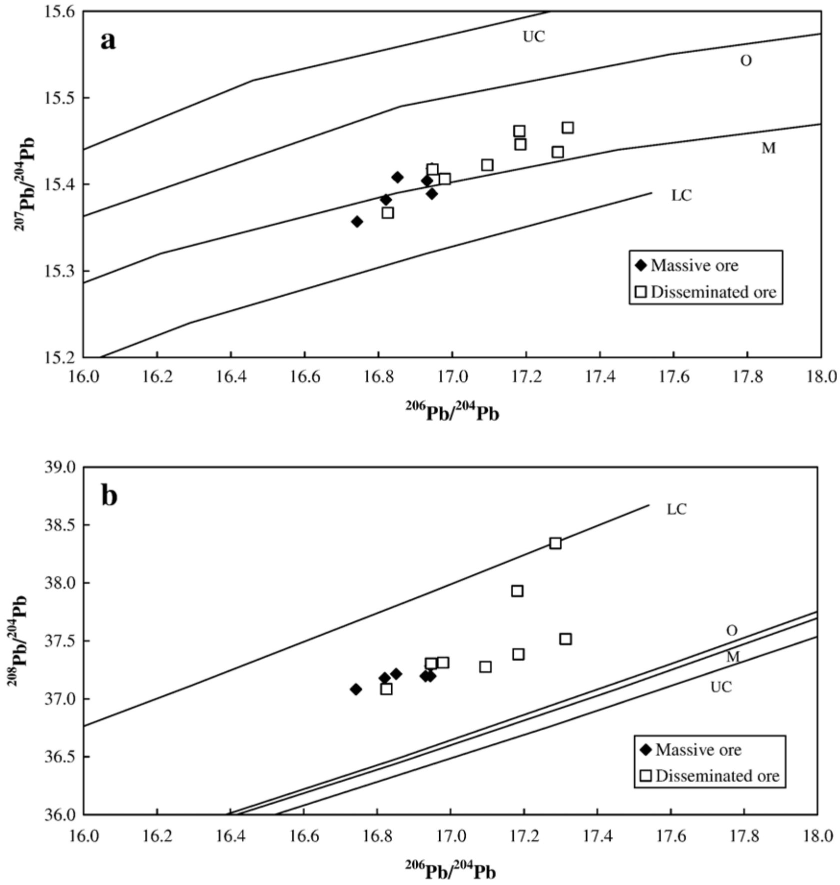

Fig. 5. $^{206}Pb/^{204}Pb$ vs. $^{207}Pb/^{204}Pb$ (a) and $^{206}Pb/^{204}Pb$ vs. $^{208}Pb/^{204}Pb$ (b) plots for the disseminated and massive ores from the Jinchuan sulfide ore deposit. The Pb isotope evolution curves are based on Zartman and Doe (1981). UC = upper crust; O = orogene; M = mantle; LC = lower crust.

828 ± 3 Ma for zircon from a dolerite dyke at Jinchuan. These U–Pb dating results are indistinguishable within analytical uncertainties from an earlier SHRIMP U–Pb age of 827 ± 8 Ma for zircon from a sulfide-bearing troctolite (Li et al., 2004). These U–Pb dates were supported by the later SHRIMP U–Pb age of about 837 Ma for zircon from a plagioclase lherzolite (Yan et al., 2005).

Baddeleyite is an ideal mineral for dating the crystallization of mafic intrusions by the U–Pb method, because it crystallizes in the late-stage, most chemically fractionated portions of the mafic magmas, it rarely occurs as xenocrysts in mafic intrusions, and it is less susceptible to Pb loss than zircon. The mineral can survive very high degrees of metamorphism but often develops a rim of polycrystalline metamorphic zircon (Heaman and LeCheminant, 1993). Many world-class magmatic Ni–Cu sulfide ore deposits have been successfully dated by the baddeleyite U–Pb method, such as Voisey's Bay (1333 ± 1 Ma; Amelin et al., 1999) and Sudbury (1850 ± 1 Ma; Krogh et al., 1984). Clearly, therefore, the baddeleyite and zircon U–Pb data indicate that the Jinchuan ultramafic intrusion and associated Ni–Cu sulfide deposit were formed at about 830 Ma (Li et al., 2004, 2005).

The Re–Os ages of the massive ores from Jinchuan (Yang et al., 2005; this work) agree within analytical uncertainties with the baddeleyite and zircon U–Pb ages. However, all other Re–Os apparent dates (Keays et al., 2004; Zhang et al., 2004; Yan et al., 2005) are older than the baddeleyite and zircon U–Pb ages, clearly indicating an open system behavior.

We note that a Sm–Nd isochron age of 1508 ± 31 Ma based on three lherzolite and plagioclase lherzolite whole rock samples and pyroxene separates (Tang et al., 1992) has been widely cited. However, the $^{147}Sm/^{144}Nd$ ratio range for the dataset is limited (0.1366 to 0.1550), and the program used for calculating the isochron age utilizes a model in which the age uncertainty was significantly reduced (Jahn, 1997). No other geochronological data, including Sm–Nd (Zhang et al., 2004; Li et al., 2005),

Table 2
Pb isotopic ratios of the disseminated and massive ores from Jinchuan

Sample number		$^{206}Pb/^{204}Pb$	$\pm 2\sigma_m$	$^{207}Pb/^{204}Pb$	$\pm 2\sigma_m$	$^{208}Pb/^{204}Pb$	$\pm 2\sigma_m$
Massive ore							
04JCII-8	Sulfide	16.932	0.009	15.404	0.008	37.195	0.020
04JCII-13	Sulfide	16.742	0.004	15.357	0.003	37.083	0.009
04JCII-13*	Silicate	16.821	0.008	15.382	0.007	37.180	0.017
04JCII-15	Sulfide	16.945	0.005	15.418	0.005	37.283	0.012
04JCII-16	Sulfide	16.852	0.007	15.408	0.007	37.215	0.018
04JCII-16*	Silicate	16.945	0.005	15.389	0.004	37.195	0.011
Disseminated ore							
04JCII-35	Sulfide	16.826	0.005	15.367	0.005	37.085	0.012
04JCII-38	Sulfide	17.095	0.002	15.422	0.002	37.275	0.006
04JCII-38*	Silicate	17.185	0.008	15.446	0.007	37.384	0.017
04JCII-39	Sulfide	16.947	0.005	15.417	0.005	37.304	0.014
04JCII-41	Sulfide	16.980	0.005	15.406	0.004	37.313	0.010
04JCII-41*	Silicate	17.314	0.008	15.465	0.007	37.515	0.018
04JCII-42	Sulfide	17.287	0.006	15.437	0.006	38.342	0.015
04JCII-43	Sulfide	17.182	0.007	15.461	0.007	37.931	0.017

The Pb isotopic ratios measured were corrected for instrumental mass fractionation of 0.10% per atomic mass unit.
*Denotes to the silicate aliquots and the others are sulfide aliquots.

Re–Os and U–Pb dates, support this date. Thus, this Sm–Nd "isochron" is very likely an errochron and the reported Sm–Nd "age" is highly problematic (Li et al., 2005).

5.2. Variation in Re–Os apparent ages of Jinchuan

Yang et al. (2005) reported a Re–Os isochron age of 833±35 Ma with an initial Os isotopic ratio of 0.279±0.018 for massive ores at Jinchuan determined by ICP-MS. Same samples were later re-analyzed by N-TIMS and yielded an age of 853±25 Ma and an initial Os isotopic ratio of 0.254±0.014 (Yang, 2005). The agreement of the two results verifies that results obtained by the two techniques in our lab are both reliable. These ages are consistent with the zircon and baddeleyite U–Pb ages and was interpreted as the formation age of the massive ore and thus, the entire deposit. Our massive ore samples yielded an apparent Re–Os isochron age of 867±75 Ma with an apparent initial $^{187}Os/^{188}Os$ of 0.235±0.027 (Fig. 3), both of which are consistent with the results given by Yang et al. (2005) and Yang (2005).

Zhang et al. (2004) analyzed eight net-textured and massive ore samples collected from No. 1 ore body for Re–Os isotopes. Seven out of the eight samples yielded an isochron age of 1034±28 Ma with an initial $^{187}Os/^{188}Os$ of 0.1503±0.0050. The authors interpreted this to be the formation age of the ores. However, this age is significantly older than the generally accepted age of the deposit based on zircon and baddeleyite U–Pb dates.

Sixteen out of nineteen net-textured ore samples, including whole-rock samples and bulk sulfide separates, reported by Keays et al. (2004) yielded an apparent isochron age of 877±34 Ma with an initial ratio of 0.163±0.011. This age is close to the zircon and baddeleyite U–Pb ages. However, the authors argued that all but three of the samples have model ages (1231–1380 Ma) that are significantly older than the 877-Ma isochron age. Therefore, they suggested that this isochron age can not be the primary age of mineralization, but rather the time at which the Jinchuan ore body was thrust from a great depth to its present position (Zhou et al., 2002), thus resetting the Re–Os isotope system. Such isotopic disturbance could be related to the addition of Re and additional radiogenic ^{187}Os to parts of the ore body by metamorphic fluids. Out of the total sixteen samples, nine samples with Re/Os ratios less than 1.3 yielded an isochron age of 1408±34 Ma and an initial $^{187}Os/^{188}Os$ ratio of 0.113±0.014. This nine-point isochron age was considered by the authors to approximate the time of mineralization of Jinchuan because it is reasonably close to the average model age and because the nine samples have low Re/Os ratio and are assumed to have been less disturbed by later addition of radiogenic ^{187}Os. However, calculation of the model age is model-dependent; it depends on the assumed initial Os isotopic ratios. Although no detailed data are available, a rough estimate suggests that the explanation made by Keays et al. (2004) can only be valid when the mantle value is taken as the initial Os isotope ratio for the model age calculation. However, such an assumption is not valid for Jinchuan because the initial Os isotopic ratio was modified by crustal contamination. If the initial $^{187}Os/^{188}Os$ ratio yielded by the sixteen point isochron (that is 0.163) is used in the calculation of the model age, most samples give model ages close to the age yielded by this isochron. Therefore, "model age" can not be used to judge if an isochron age is geologically meaningful, and the apparent age of 877±34 Ma yielded by the sixteen point isochron is adopted in our data compilation.

Eleven net-textured samples collected by Yan et al. (2005) yielded an apparent isochron age of 999±140 Ma with an initial Os ratio of 0.159±0.047. This apparent age is close to, but older than, the zircon and baddeleyite U–Pb ages. However, the authors suggested that this apparent Re–Os isochron age represents later tectonic-hydrothermal events based on the fact that the initial Os isotope ratios obtained is higher than the mantle value. They used the supposed age of a dike cutting the ore body (1336 Ma) to argue that mineralization of the Jinchuan ore deposit must have occurred prior to this time. Unfortunately, however, the authors neither cited earlier literature nor presented the analytical method and original data to support this age. To our knowledge there is no such date on the ore deposit, instead, the age was derived from rock of the Qilian Mountain (Tang and Li, 1995). Consequently, this age can not be used to constrain the formation time of the Jinchuan deposit.

We obtained apparent Re–Os ages of 1117±67 Ma and 1074±120 Ma and initial Os ratios of 0.120±0.012 and 0.162±0.017, respectively, for the disseminated ores and sulfides from the disseminated ores of Jinchuan (this work, Section 4.1).

By integrating available apparent Re–Os ages, we found an important correlation between apparent Re–Os isochron ages and initial Os isotope ratios on the one hand, and the ore types on the other (Fig. 6). Massive ores yield the youngest apparent Re–Os isochron ages (853±25 Ma; Yang, 2005) and 867±75 Ma (this work) as well as the highest initial Os isotopic ratios of 0.254±0.014 (Yang, 2005) and 0.235±0.027 (this work). These ages are consistent with the zircon and baddeleyite U–Pb ages which are taken as the formation age of the deposit. Net-textured ores yield an apparent Re–Os age older than that of the massive ores (1043±28 Ma, Zhang et al., 2004) or overlapping with that of the massive ores (877±34 Ma, Keays et al., 2004). The net-textured ores also have lower initial Os ratios than the massive ores (0.150±0.005, Zhang et al., 2004; 0.163±0.011, Keays et al., 2004). Disseminated ores yield the oldest apparent age (1117±67 Ma; this work) although it partially overlaps with that of some of the net-textured ores. Meanwhile, these samples yield the initial Os ratio which is the lowest (0.120±0.012; this work). Sulfides from disseminated ores yield an apparent age (1074±120 Ma; this work) agreeing with that of the disseminated ores. These samples yield the initial Os ratio (0.162±0.017; this work) which agrees with those of the net-textured ores. Thus, sulfides from disseminated ores show Re–Os apparent age and initial Os ratio intermediate between those of the disseminated and the net-textured ores. It is worth to note that samples II-1-1 to 6 contain more sulfide minerals (6–8 modal%) than that of disseminated ore samples JCII-035-043 (less than 2 modal%) and samples containing more sulfides give higher initial Os isotopic ratio. Obviously, apparent Re–Os isochron ages and initial Os isotopic ratios are correlated with the textures of the ores, or the contents of sulfides in the ores (Fig. 6).

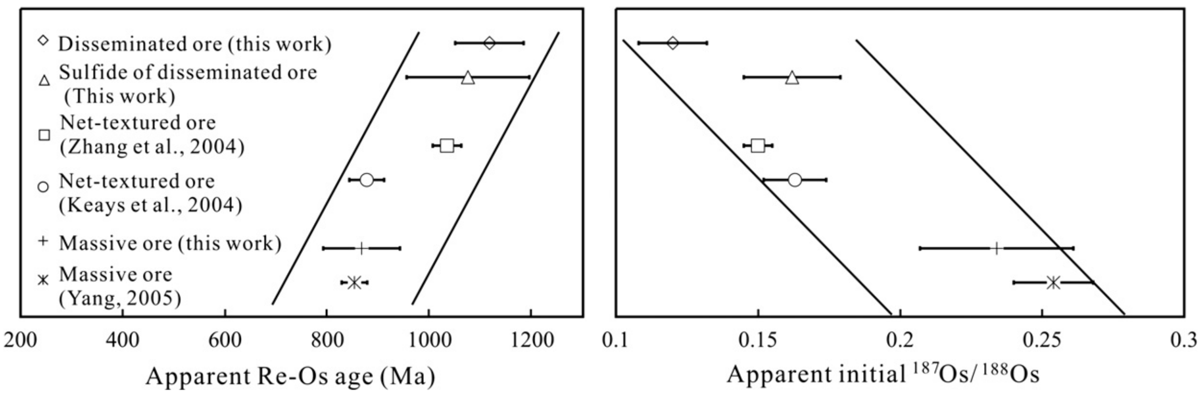

Fig. 6. Trends of apparent Re–Os ages (a) and initial $^{187}Os/^{188}Os$ ratios (b) of different types of ores from Jinchuan (data are from Keays et al., 2004, Zhang et al., 2004, Yang et al., 2005 and this work).

5.3. Previous model

Many previous workers have used the later disturbance model to explain the inconsistent apparent Re–Os ages of Jinchuan. They suggest that the deposit was formed in the Mesoproterozoic and suffered later disturbance(s) that reset the primary Re–Os isotope system. This would cause the isochron to rotate in a clockwise direction in the $^{187}Re/^{188}Os$–$^{187}Os/^{188}Os$ plot, thereby yielding a younger apparent age and a higher apparent initial Os ratio (Keays et al., 2004; Yan et al., 2005).

This model has successfully explained apparent Re–Os ages in some sulfide magmatic ore deposits, such as Voisey's Bay (Lambert et al., 2000) where the later disturbance model is supported by additional geochronological and geological evidence. However, this model is unreasonable for Jinchuan for several reasons.

First, there is no evidence of multi-stage magmatism at Jinchuan. Contacts between different rock types are gradational and no clear divisions have been observed in the field by previous workers (e.g. Chai and Naldrett, 1992) or by us, although three intrusive stages were identified by Tang and Li (1995). The roughly symmetrical distribution of the different rock types in most parts of the deposit does not favor multi-stage magmatic injection because later intrusions would likely be injected along weak zones, such as the walls of the earlier intrusion, rather than in the middle of an earlier magmatic body (Lu, 1993). Tang and Li (1995) assigned the dunite to their third stage of intrusion, but the rock contains 70–90 modal% olivine, and thus must have been rigid. It could not, therefore, flow and be "injected" to the center of the "second stage" lherzolite (Lu, 1993). Thus, the concept of multi-stage magmatism at Jinchuan needs a critical scrutiny although this possibility can not be ruled out entirely. Small-scale later injection did take place at Jinchuan, as shown by the presence of vein-type massive ore. However, Re–Os dating results suggest the time span between injection of the massive ore and the formation of the main intrusion is indistinguishable.

Secondly, if later thermal or tectonic events affected the Jinchuan deposit (Keays et al., 2004; Yan et al., 2005), they should be younger than the formation age of the intrusion represented by the zircon and baddeleyite U–Pb ages. However, the youngest apparent Re–Os ages are consistent with the zircon and baddeleyite U–Pb ages. Thus, the available apparent ages can not be explained as the age of later metamorphic events.

Thirdly, all apparent ages older than the formation of the intrusion are not reliable or are based on invalid assumptions as discussed in the previous sections.

Based on these consideration, we conclude that the later disturbance model is not applicable to the Jinchuan deposit.

Lambert et al. (1998) proposed a model which relates Re–Os concentrations, Os isotopic ratios, crustal contamination and the *R* factor (*R* is defined as the effective mass of silicate magma with which a given mass of sulfide magma has equilibrated). This model suggests that ores experienced a high *R* factor process would give relatively homogeneous Os isotopic initial ratios, making it possible to obtain a reliable Re–Os age. Those ores, such as the ones at Duluth, Sudbury and Stillwater, which experienced a relatively low *R* factor process, would give high and heterogeneous initial ratios suggesting that a reliable Re–Os age could not be obtained. However, this model does not explain why ores with different textures yield different apparent Re–Os isochron ages at Jinchuan. Some additional process, very likely the diffusion of Os in sulfide minerals after segregation of the sulfide droplets and *R* factor process, must have operated to produce the observed trend at Jinchuan.

5.4. Os diffusion in ores with different textures

The most important factor constraining the diffusion of elements and isotopes is the diffusion coefficient of the element. Experiments conducted by Brenan et al. (2000) give a closure temperature for Os diffusion of about 400 °C in pyrrhotite having a grain size of 1 mm, but indicated a closure temperature of more than 500 °C for pyrite. No diffusion parameters are available for other sulfide and opaque minerals. However, pentlandite, pyrrhotite, chalcopyrite and magnetite form a Re–Os isochron which yielded an apparent age that is consistent with the disturbance time in the Voisey's Bay sulfide deposit (Lambert et al., 2000). This suggests that Os isotope equilibrium was achieved for those minerals when the later disturbance took place, suggesting that pentlandite, chalcopyrite and magnetite have closure temperatures for Os diffusion similar to that of pyrrhotite. Temperature is also an important factor for diffusion. Os diffusion rate in sulfides at high temperature is several orders of magnitude faster than that of 400 °C. For example, only about several hundreds to several thousands of years are required for the core of a pyrrhotite with a grain size of 0.5 mm to begin to be affected by diffusive exchange with an external Os source at temperatures of 600 °C and 500 °C (Brenan et al., 2000).

However, Burton et al. (1999) reported that sulfide inclusions in silicate minerals from a spinel lherzolite from New Mexico preserved low Os isotopic ratios, suggesting that they have been shielded from reaction or diffusion by their silicate hosts, that the sulfide diffusion through silicate minerals is impaired by the high partition coefficient for Os between sulfide and silicate phases and that silicate minerals act as a barrier for Os diffusion even at mantle temperatures.

On the other hand, ore textures are related to the area of the cross section through which Os diffuses between different minerals. Based on microscopic observation, a simplified model for the ore textures is postulated (Fig. 7). In the disseminated ores with less than 5 modal% sulfides, the ore minerals are separated from each other by the dominant silicate minerals (Fig. 7). Os can not diffuse and be exchanged between disseminated sulfide minerals even at the magma temperatures because of the presence of the barrier of the Os diffusion, the dominant silicate minerals. In the massive ores consisting entirely of sulfide minerals (Fig. 7), the sulfides are closely connected and Os can diffuse and be exchanged between these minerals quickly at high temperature so that Os isotopic equilibrium can be achieved. The net-textured ores contain about 25 modal% opaque minerals and these minerals are partly connected in a 'net-like' fashion (Fig. 7). Os diffusion is impaired by limited connection surfaces between opaque minerals even at high temperatures and Os can only diffuse and be exchanged between these minerals to a limited extent. Clearly, Os isotope equilibrium would be achieved faster in massive ores than in the net-textured ores where the opaque minerals are only partially connected, and little or no Os diffusion would occur in disseminated ores. Other than those, the greater the amount of Os being diffused is, the longer it will take for isotope equilibrium to be achieved.

5.5. Post-segregation-(deposition) Os diffusion model

Based on the diffusion behavior of Os in ores with different textures, we propose a model to explain the conditions under which heterogeneous initial Os isotope ratios can be maintained and how homogeneous initial ratios could be achieved, thereby clarifying the geologic meaning of the apparent Re–Os isochron ages.

Suppose that the deposit was formed at time t_0 and Os of the deposit was derived from a mixed source (mantle-derived and crust-derived ones). Os isotope ratios in different parts of the deposit were different at time t_0 because of mixing and the R factor process. In

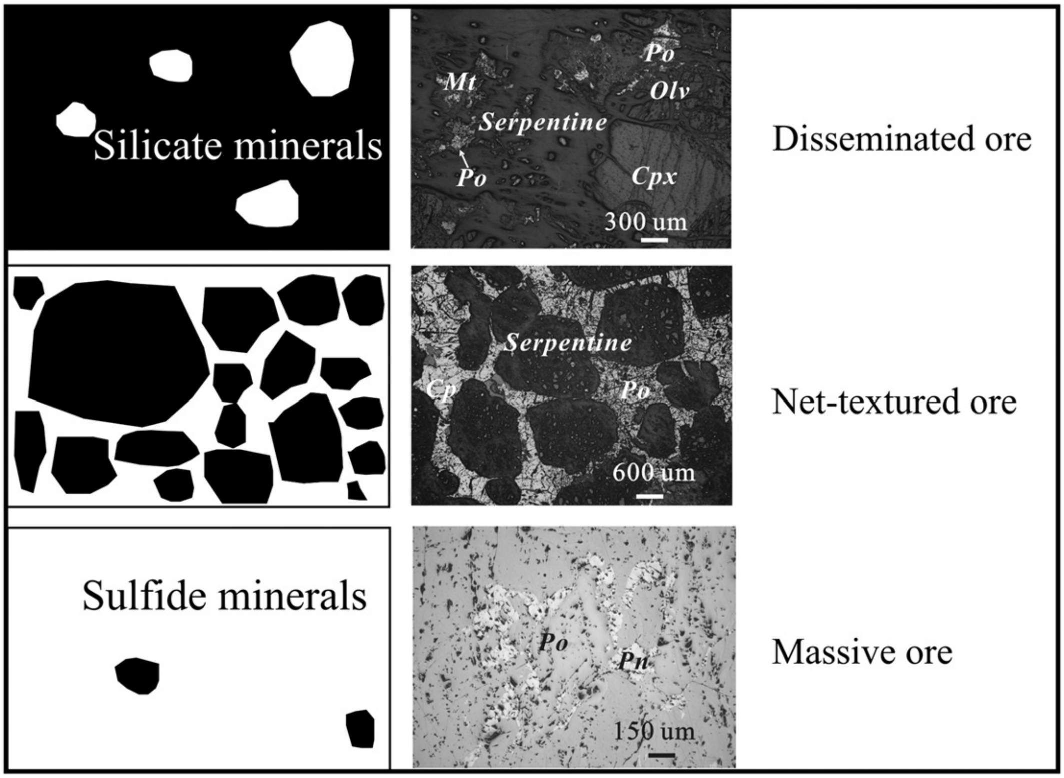

Fig. 7. A sketch of the ore textures of the Jinchuan sulfide ore deposit (left) with microphotographs of representative ore samples (right; disseminated ore, JCII-035, net-textured ore, JCII-057 and massive ore, JCII-008, reflected, plane-polarized light).

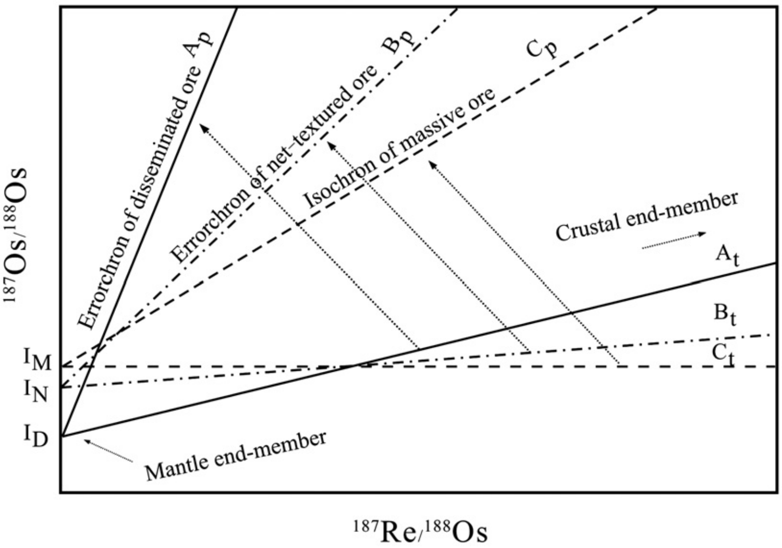

Fig. 8. Os isotopic evolution of different types of ores based on the post-segregation-(deposition) Os diffusion model.

fact, the R factor process is essentially a mixing process operating in the dynamic mafic magma system. Re–Os isotopic data of the segregated sulfide droplets would yield a mixing line in a $^{187}Re/^{188}Os$ vs. $^{187}Os/^{188}Os$ plot (A_t in Fig. 8). The low $^{187}Re/^{188}Os$ end of the mixing line gives the $^{187}Os/^{188}Os$ ratio of the end-member of the mantle-derived mafic magma (I_D in Fig. 8). The crustal end-member possessed high $^{187}Re/^{188}Os$ and high $^{187}Os/^{188}Os$ ratios. Heterogeneous initial Os isotope ratios would be maintained in disseminated ores because Os could not diffuse between the opaque minerals which were isolated by the silicate minerals. Because of radioactive decay since time t_0 to the present, different parts of the ore would accumulate different amounts of the radiogenic daughter isotope (^{187}Os) according to the $^{187}Re/^{188}Os$ ratios of the different parts. An "isochron" would be produced by adding radiogenic isotopes to the heterogeneous "initial values". Such an isochron (A_P of Fig. 8) is obviously an errochron which yields an age that is older than the formation age and is geologically meaningless because the initial ratios were not the same and the prerequisite of radioactive dating is not met. However, the initial isotope ratio yielded by this errochron (A_p) approximately equals the mantle value (I_D in Fig. 8).

In contrast, the Re and Os isotopes will evolve differently in massive ore. Os isotopes can easily diffuse and exchange among the sulfide minerals during cooling of the ore because of low closure temperatures of Os diffusion and the close connection among these minerals. A homogeneous initial Os isotope composition (I_M in Fig. 8) for different parts of the ore would soon develop because of this fast diffusion. The time interval needed for the re-equilibration to be achieved would be about 1 m.y. at temperatures of about 400 °C and is not detectable by the present geochronological method. This time interval would be several orders of magnitudes shorter if the temperature was high (Brenan et al., 2000). The homogeneous initial ratios give a horizontal line in the $^{187}Re/^{188}Os$ vs. $^{187}Os/^{188}Os$ plot (C_t; Fig. 8). An isochron (C_P in Fig. 8) would be formed after radioactive decay from time t_0 to the present. This isochron yields an age corresponding to the age of mineralization, but an initial Os isotope ratio which is higher than the mantle value and meaningless (I_M in Fig. 8).

Re–Os isotopes of the net-textured ore behave in a fashion intermediate between the disseminated and massive ores. Partial re-equilibration could be achieved during cooling of the ore because the diffusion of Os is faster than in the disseminated ore, but slower than in the massive ore. Different degrees of Os isotope re-equilibration would be achieved and different degrees of heterogeneity in initial Os isotope ratios maintained, producing line B_t between A_t and C_t in the $^{187}Re/^{188}Os$ vs. $^{187}Os/^{188}Os$ plot (Fig. 8). This line points to an "initial $^{187}Os/^{188}Os$ ratio (I_N in Fig. 8) that is intermediate between I_M and I_D and is geologically meaningless. Starting from heterogeneous "initial values" an errochron (B_P in Fig. 8) would be formed. This errochron yields an apparent age that is older than that of the mineralization but younger than the apparent age given by the errochron yielded by the disseminated ore and is also geologically meaningless. It may yield an apparent age that equals the formation time of the deposit but with large uncertainty due to heterogeneity in the initial Os isotope ratio and thus the resulting initial (I_N) is also meaningless.

5.6. Implications for the Jinchuan sulfide deposit

In order to use the model to explain the apparent age-ore texture trend observed at Jinchuan, one of the key issues is whether or not the Os in the Jinchuan deposit was derived from a mixed source. The following lines of evidences suggest that crustal contamination occurred in the Jinchuan magma.

Six out of seven disseminated ore samples reported in this study show a correlation between the initial $^{187}Os/^{188}Os$(830 Ma) ratio and the reciprocal of Os concentration (Fig. 9a). Such a linear correlation is often used to argue for a mixing relation (Faure and Mensing, 2005). A similar correlation is shown by the net-textured ores reported by Yan et al. (2005) (Fig. 9b) and Zhang et al. (2004). Data shown in Fig. 9a also suggest that Os concentrations vary widely in different parts of the ore body (04JCII-35 and the others), however, they plot on the same isochron. This may be caused by segregation and accumulation of droplets of Os minerals and Os-bearing sulfide minerals. The Jinchuan ultramafic rocks all exhibit negative $\varepsilon_{Nd}(T)$ values that decrease with increasing La/Sm ratios, suggesting that their parent magmas were derived from an enriched lithospheric mantle and experienced crustal contamination (Li et al., 2005). Trace element concentrations of ultramafic rocks from Jinchuan show primitive mantle-normalized patterns with negative Nb–Ta anomalies, which are consistent with derivation from mantle magmas variably contaminated by crustal materials (Song et al., 2006).

The sulfur isotopic composition of the Jinchuan ores might be used to argue against crustal contamination of the parental magmas because the $\delta^{34}S$ values are constant, generally in the range of −2‰ to +8‰, with most values between −2‰ to +2‰ (Tang and Li, 1995; Ripley et al., 2005). However, if the crustal contaminant is magmatic in origin, or is Archaean in age, such materials would normally possess near zero $\delta^{34}S$ values. If this is the case, S isotopes can not be used as a good indictor of crustal contamination. Nevertheless, further investigation is required to identify the real crustal contaminant at Jinchuan.

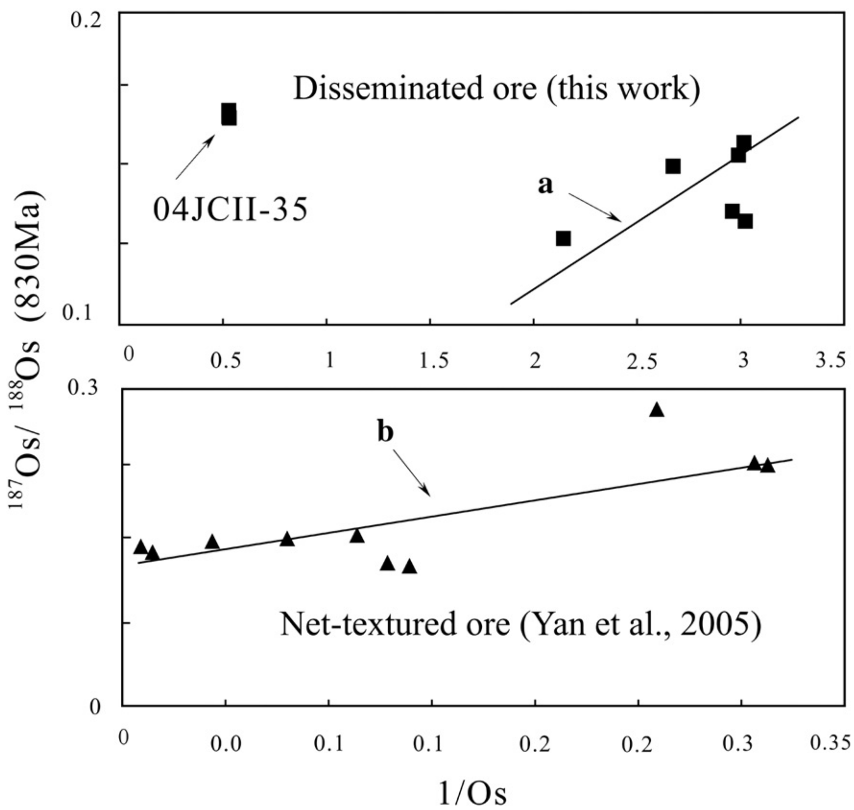

Fig. 9. Correlation of the initial $^{187}Os/^{188}Os$ ratios and the reciprocals of the osmium concentrations. a) Disseminated ore (this work); b) net-textured ore (Yan et al., 2005).

Another important issue is the temperature of sulfide (and magnetite) precipitation and estimated cooling rates for the magma and/or sulfide phases in Jinchuan. Unfortunately, no precise information on the cooling rate has been found from the literatures and only a rough estimate can be made. Sulfide segregation, producing layering of the pure sulfide, sulfide rich crystal mush and sulfide poor magma, took place when the sulfide droplets were still at the molten state (Tang and Li, 1995; De Wall et al., 2004), that is, at a temperature higher than 850 °C (Stone et al., 1989). At such a high temperature environment, Os diffusion was rapid. Take an extreme cooling rate of 1000 °C/Ma, it required about 0.4 Ma for the system to cool from 850 °C to 400 °C. Such a time interval is long enough for Os isotopic re-equilibrium to be achieved, especially at high temperatures, if mineral grains are connected each other closely.

Sulfide saturation was achieved in a deep magma chamber at Jinchuan because of contamination of the primary, high-Mg basaltic magma (Chai and Naldrett, 1992) by old radiogenic crustal material. Sulfide droplets with heterogeneous $^{187}Os/^{188}Os$ ratios were segregated from the contaminated magma. An *R* factor process followed and Os isotopic compositions of the sulfide droplets continuously changed as the *R* factor process proceeded, but the heterogeneity of the Os isotope ratios remained in different sulfide droplets to different degrees since the *R* factor for Jinchuan varies from about 1000 to 50000 (Yang, 2006). The droplets then accumulated in different parts of the deposit, forming different types of ores with different Re and Os concentrations and Os isotope ratios gained in the *R* factor process. Os in different types of ores evolved in different ways as described above. Thus, the errochron generated by the disseminated ore yields a geologically meaningless apparent Re–Os age and an initial Os isotopic ratio which approximates the mantle value. In contrast, the net-textured ore yields both geologically meaningless apparent ages and initial ratios, whereas the massive ore yields an apparent Re–Os age which corresponds to the formation time of the intrusion and an initial ratio which is geologically meaningless. Additionally, monosulfide solid solution crystallization that affects Re–Os fractionation in magmatic sulfide ore systems played a role in generation of the variable Re/Os ratios in different parts of the ore, which is a prerequisite to yield a precise isochron (Brenan, 2002).

Being chalcophile elements, Pb and Os may both concentrated in sulfide minerals, thus, Pb isotopic compositions may provide an independent check for the Os isotopic behavior predicted by the proposed model. The determined Pb isotopic ratios can be treated as the initial ratios because that U and Th are very low in sulfides. Heterogeneous Pb isotopic ratios in the disseminated ores suggest that initial Pb isotopic ratios were heterogeneous and that Pb isotopic equilibrium was not achieved during cooling of the ore. Heterogeneity in initial Pb ratios is probably caused by the contamination of the crustal material although the end-members could not be strictly constrained. Pb could not diffuse between disseminated sulfide minerals because dominant silicate minerals separate sulfides and Pb is strongly partitioned into sulfides. Thus, a later re-equilibration was not achieved. In contrary, the homogeneous Pb isotopic ratios in the massive ore suggest that equilibrium of Pb isotopes was achieved during cooling of these ores because Pb could easily diffuse between closely connected sulfide minerals. Similar diffusion behavior of Pb and Os supports our model.

Our model is capable of explaining the relation between the apparent Re–Os ages, the initial Os isotope ratios and the ore-types at Jinchuan. However, data provided by Keays et al. (2004) cannot be straightforwardly explained by this model. The explanation of their data may be related to different sampling locations or cooling rates, different ore textures, different Re–Os concentrations and different Re/Os ratios, all of which may be related to the time required for re-equilibration of the Os. The rate of re-equilibration may also be related to variable *R* factors, which determine the variations of the Os isotopic ratios immediately after segregation of the sulfides. A thorough discussion cannot be made before more detailed information becomes available.

5.7. Implications for Re–Os dating of the other sulfide deposits

Based on a review of most major magmatic Ni–Cu sulfide deposits in the world, including those at Noril'sk, Sudbury, Jinchuan, Voisey's Bay and Kambalda, Naldrett (1997, 1999) concluded that interaction of the source magma with country rock is one of the key factors for the genesis of world class Ni–Cu–PGE deposits. Heterogeneous initial Os isotopic ratios would arise because of crustal contamination and R factor processes (Lambert et al., 1998), partly explaining why apparent Re–Os ages of some sulfide ore deposits are not consistent with the formation age of the host intrusions. The model proposed in this work is applicable to the Jinchuan sulfide ore deposit, however, its applicability to other sulfide deposits depends on whether the state of equilibrium of initial Os isotopes can be achieved, and on the post-segregation–(deposition) diffusion of Os during cooling of the ores. Whether re-equilibration of the initial Os isotopic ratios can be achieved further depends on several factors which affect the diffusion of Os in ores.

Variation of the Os isotopic composition immediately after segregation of sulfides is the first issue which should be considered. The larger the contrast of the Os isotopic compositions of the parent magma and the crustal contaminant, the larger the heterogeneity of Os isotopic ratios of the sulfide droplets segregated from the silicate magma. The R factor is an even more important factor. The lower the R factor, the lower the Os concentration, the higher the γ_{Os} value (defined as the percentage difference between the Os isotopic composition of a sample and the average chondritic composition for a specific time) and the larger the variability of the γ_{Os} values (Lambert et al., 1998). Thus, ores that experienced relatively low R factor processes would show a larger range of Os isotopic ratios and have more difficulty re-equilibrating during cooling of the ores. In contrast, ores that experienced very high R factor processes (>50000) would show homogeneous initial Os isotopic ratios, such as at Kambalda and the J–M reef in the Stillwater Complex (Lambert et al., 1998). As discussed in previous sections, closure temperatures of Os diffusion in ore-forming minerals are an important factor. Pentlandite, pyrrhotite, chalcopyrite and magnetite show low closure temperatures for Os diffusion, whereas pyrite has a relatively high closure temperature. Unfortunately, the closure temperatures of Os diffusion in other Os-bearing minerals are unknown. The textures and Os concentrations of ores and minerals are also important. Concentrations of Os vary for different deposits and different parts of a single deposit, and thus the quantity of Os to be exchanged and the time required for equilibrium of Os isotopes to be achieved are different. In addition, the cooling rate should be taken into account; a prolonged high-temperature environment facilitates re-equilibration of Os during slow cooling, whereas re-equilibration of Os isotopes among sulfide minerals could not be achieved during fast cooling.

The behavior of the Re–Os isotopic system of chromite from the Peridotite Zone of the Stillwater Complex, Montana, US is somewhat similar to that of the Jinchuan ultramafic complex, although the diffusion coefficient of the chromite is not known. The regression line of the samples from unit H, consisting of chromite-bearing samples (1 to 6 modal% chromite in the disseminated ore and 13 to 30% and 70% in the net-textured ore), yields an age of 2904±43 Ma, which is older than the 2700 Ma formation age of the complex (Horan et al., 2001). This gives an initial Os isotope ratio of 0.11160±0.00044, equivalent to γ_{Os} of 2.20±0.44 (calculated in this study) which is close to the mantle value. The regression line of unit G samples, containing 40 to 70 modal% chromite (net-textured ore) and about 94 modal% chromite (massive ore), yields an age of 2771±120 Ma, which is consistent with the formation age of the complex (Horan et al., 2001) and an initial Os isotope ratio of 0.11329±0.00059, equivalent to γ_{Os} of 3.75±0.54 (calculated in this study) which is significantly higher than that yielded by the unit H samples. Variation in γ_{Os} values by calculation of single samples gives a similar contrast (Horan et al., 2001). The Re–Os apparent ages and initial Os isotopic ratios shown by units H and G are consistent with the model proposed in this study; that is, the apparent age yielded by ores containing less Os-bearing minerals is older than the formation age of the intrusion, but, with an initial ratio close to the mantle value. In contrast, the apparent age given by ores containing more abundant or massive Os-bearing minerals is approximately equal to the formation age of the intrusion, but with a large uncertainty and the initial ratio is higher than the mantle value. As such, the model proposed in this study provides an alternative or additional explanation for the dataset. Equilibration of the Os isotopic composition was not achieved in units A and B (Horan et al., 2001).

At the Babbitt sulfide deposit associated with the Duluth complex three out of four massive ore samples containing 100 modal% sulfide minerals (pyrrhotite, chalcopyrite, cubanite, pentlandite and locally talnakhite) yield an isochron corresponding to an age of 1580 Ma, which is intermediate between the formation age of the Duluth Complex (1100 Ma) and the age of the Virginia Formation (1851 Ma) which is considered to be the principal source of the crustal contaminant (Ripley et al., 1998). This suggests that a partial re-equilibration was achieved in the massive ores, probably through post-segregation diffusion of Os among the sulfide minerals. However, a full re-equilibrium of Os isotopes was not achieved, suggesting either fast cooling of the ore, or, low R factors (<1000; Ripley et al., 1998), or a combination of these and other unknown factors.

Although the model proposed in this study explains the variation of the apparent Re–Os ages and initial Os isotope ratios with textures of the ores at Jinchuan and partially explains Re–Os isotopic behaviors in a few other sulfide ore deposits in the world, the applicability of the model to other sulfide deposits depends on several factors. However, our model can at least provide additional information on the behavior of Os isotopes during mineralization.

6. Conclusions

By comparing apparent Re–Os isochron ages of disseminated and massive ores as well as sulfide from the disseminated ore obtained during this study and previous Re–Os dates of the Jinchuan sulfide ore deposit, we found that there is a correlation among apparent Re–Os ages, initial Os isotope ratios and the ore textures. Massive ores consisting entirely of sulfide minerals yield the youngest apparent Re–Os ages, which are consistent with the formation age of the intrusion determined by zircon and baddeleyite U–Pb dating, and geologically meaningless high initial Os isotopic ratios. Net-textured ores containing medium amounts of opaque minerals yield geologically meaningless apparent Re–Os age and initial isotope ratios. Disseminated ores containing only a few percent of opaque minerals yield the oldest but geologically meaningless apparent Re–Os age and initial Os isotopic ratios which approximate the mantle value. Because the Os diffusivities in pentlandite, pyrrhotite, chalcopyrite and magnetite are high at temperatures higher than 400 °C and silicate minerals behave as the barrier of Os diffusion even at the mantle temperatures, we propose that post-segregation-(deposition) diffusion of Os played an important role in homogenization of Os isotopic ratios after segregation of sulfide droplets from the mantle derived magma, which was contaminated by crustal material and affected by *R* factor processes. The model explains the trends observed in Jinchuan and partially explains datasets from a few other sulfide deposits. An implication of the model is that massive ore of the magmatic sulfide ore deposit may be the most suitable samples for Re–Os dating of the deposit. However, the applicability of the model to other sulfide deposits depends on many factors affecting equilibrium of Os isotopic compositions in the ores.

Acknowledgements

We thank Liu Tongyou, Qiao Fugui, Jiang Jiapu, Lu Yaojun, Yin Maohong and Liu Lei of the Jinchuan Nonferrous Metal Cooperation for assistance with the fieldwork, Li Xianhua of the Guangzhou Institute of Geochemistry, Chinese Academy of Sciences, Tang Zongli, Yan Haiqing, Qian Zhuangzhi and Jiao Jiangang of Chang'an University, Li Chusi of Indiana University, Su Shangguo of China University of Geosciences, Beijing, for enlightening discussions and Dr. Paul Robinson of Dalhousie University, Canada for an English editing. This work is supported by the NSFC (grants 40373010 and 40534020). We also thank two anonymous reviewers and the editor, Roberta Rudnick for offering comments which have substantially improved the manuscript.

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